

Retail Sales & Customer Analytics Dashboard

Python + SQL Analysis Report

Portfolio Project: retail-analytics-project-full_enterprice_level
Prepared for Power BI dashboard development

This report summarizes the completed Python and PostgreSQL analysis phase. It validates core business KPIs, documents SQL outputs, and converts raw retail data into dashboard-ready business insights.

1. Project Overview

Objective: Build an enterprise-style Retail Sales & Customer Analytics Dashboard using Python, PostgreSQL, and Power BI.

Tech stack: VS Code, Python 3.13.9, venv, pandas, numpy, matplotlib, seaborn, PostgreSQL, pgAdmin 4, and Power BI.

Data source: Synthetic retail dataset containing customers, products, stores, orders, and returns.

Dataset Summary

File	Rows	Purpose
customers.csv	50000	Customer profile and segment data
products.csv	1200	Product, brand, category and pricing data
stores.csv	120	Store location and type data
orders.csv	300000	Transaction-level sales data
returns.csv	16513	Returned order and refund data

2. Data Pipeline Summary

Python phase: Raw CSV files were read, validated, cleaned, and transformed. Date columns were converted, order date features were created, and analysis outputs were saved for dashboard use.

SQL phase: Raw CSV files were imported into PostgreSQL tables because the SQL schema matched the raw file structure. SQL KPIs were validated against Python outputs.

Power BI phase: The next step is to use processed analysis outputs and/or PostgreSQL tables to build interactive dashboard pages.

3. KPI Validation

Executive KPI Output

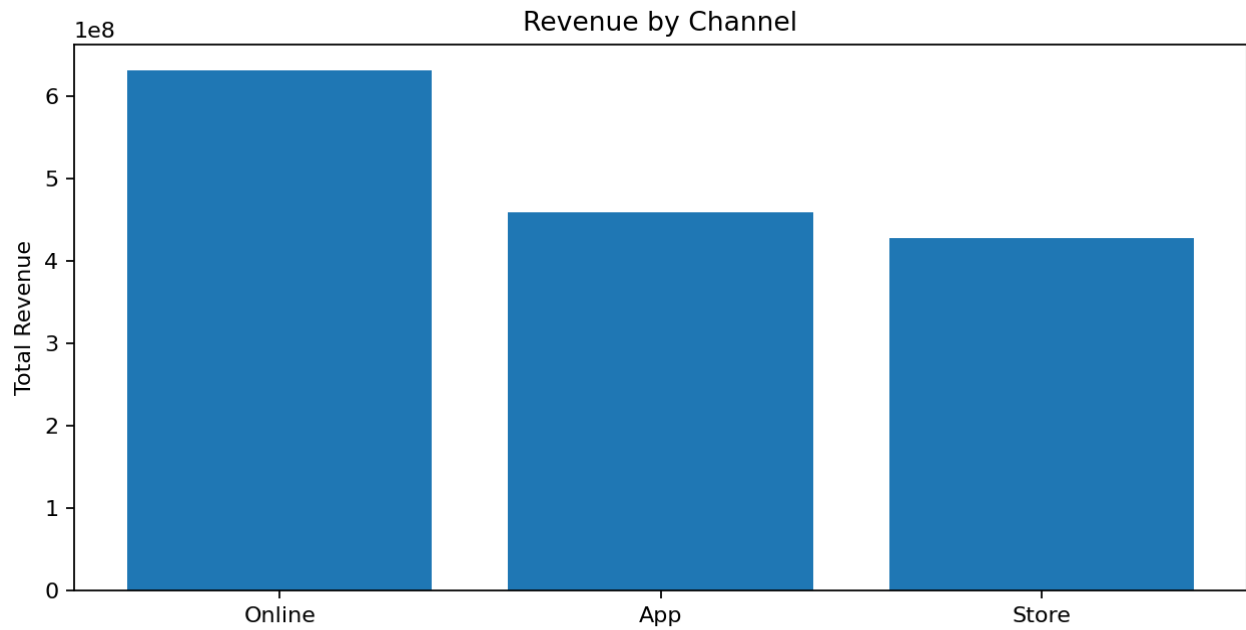
Metric	Value
Total Revenue	Rs. 1,517,017,257.00
Total Profit	Rs. 404,822,271.44
Total Orders	300,000
Total Customers	49,887
Total Quantity Sold	527,126
Profit Margin %	26.69%

Insight: Python and SQL outputs matched exactly for the main KPI query. This confirms that the PostgreSQL import and Python transformation logic are aligned.

4. Sales Channel Analysis

Revenue by Channel

channel	total_revenue	total_orders	avg_order_value
Online	Rs. 631,054,446.86	125,466	Rs. 5,029.68
App	Rs. 458,828,240.67	90,305	Rs. 5,080.87
Store	Rs. 427,134,569.47	84,229	Rs. 5,071.11

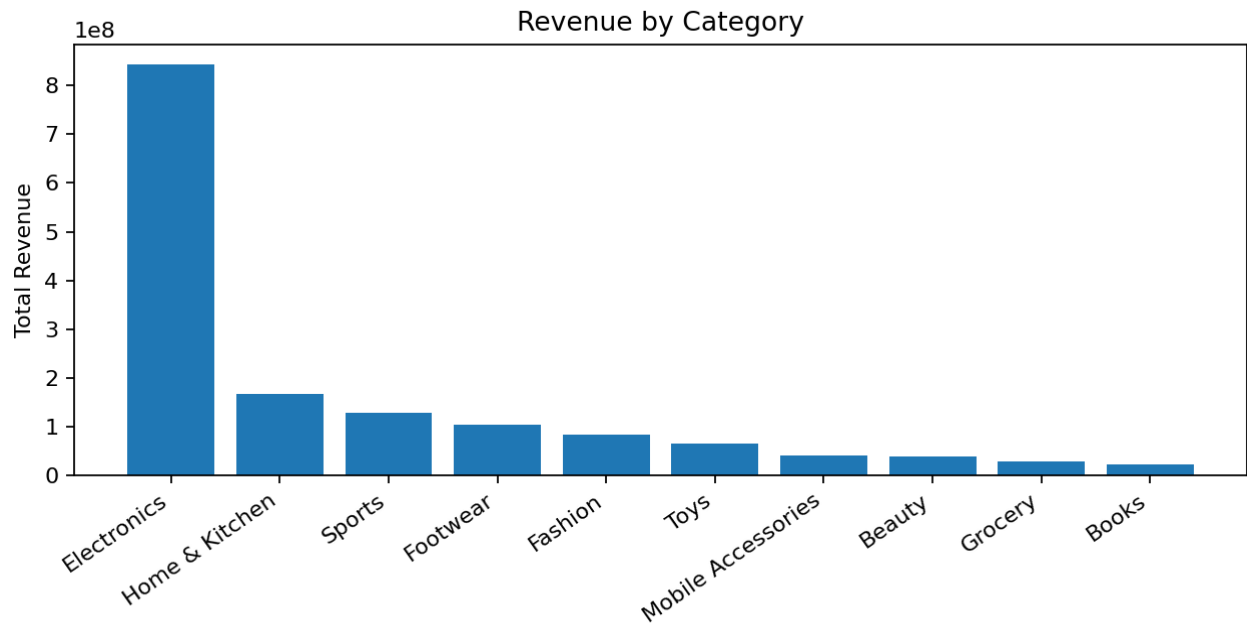


Insight: Online generated the highest revenue because it had the highest number of orders. App had the highest average order value, showing stronger order value per transaction.

5. Product Category Analysis

Revenue and Profit by Category

category	total_revenue	total_profit	total_orders	total_quantity_sold	profit_margin_percentage
Electronics	Rs. 842,627,649.69	Rs. 224,685,867.17	29,651	51,755	26.66%
Home & Kitchen	Rs. 166,254,026.23	Rs. 43,653,951.11	28,240	49,525	26.26%
Sports	Rs. 127,811,528.81	Rs. 34,793,613.68	31,173	55,011	27.22%
Footwear	Rs. 104,034,550.71	Rs. 27,435,612.97	30,023	52,998	26.37%
Fashion	Rs. 82,999,716.20	Rs. 22,102,647.86	29,660	52,023	26.63%
Toys	Rs. 65,665,717.88	Rs. 18,267,051.60	35,462	62,381	27.82%
Mobile Accessories	Rs. 40,493,508.64	Rs. 10,874,024.05	30,999	54,376	26.85%
Beauty	Rs. 37,821,335.15	Rs. 9,817,173.82	26,630	46,847	25.96%
Grocery	Rs. 27,503,482.47	Rs. 7,240,992.46	28,418	49,724	26.33%
Books	Rs. 21,805,741.22	Rs. 5,951,336.72	29,744	52,486	27.29%

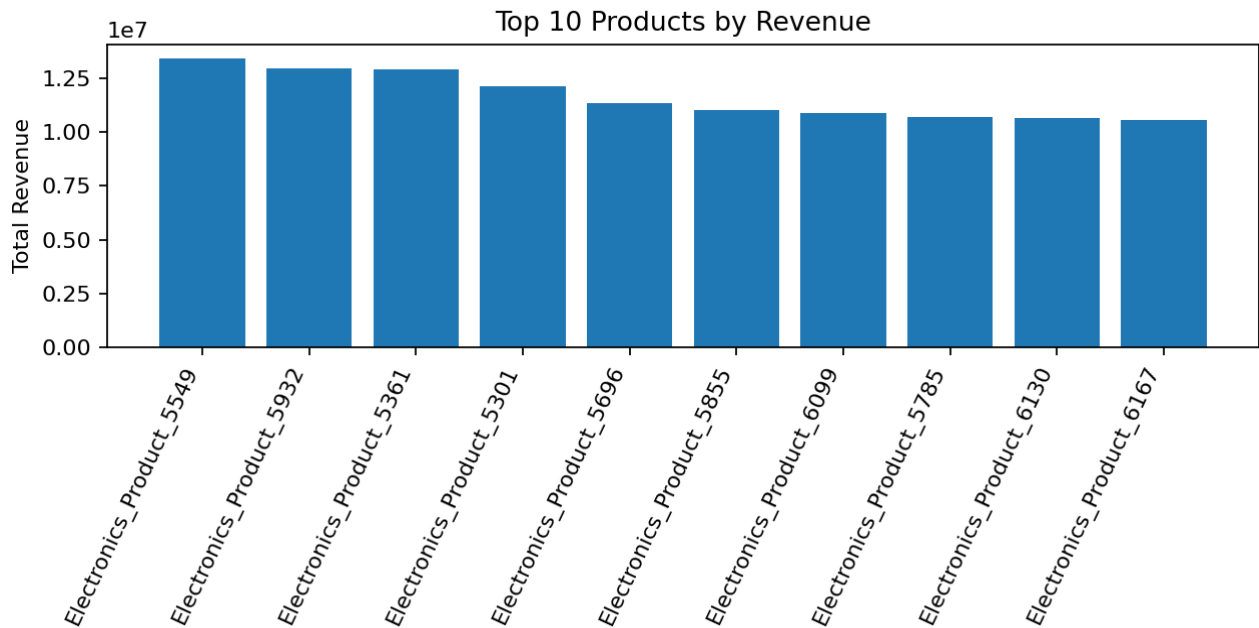


Insight: Electronics and Home & Kitchen are the leading revenue categories. Category performance can guide inventory planning, marketing focus, and category-level profitability decisions.

6. Top Product Analysis

Top 10 Products by Revenue

product_name	category	total_revenue	total_profit	total_orders	total_quantity_sold	profit_margin_percentage
Electronics_Product_5549	Electronics	Rs. 13,403,543.85	Rs. 2,121,420.37	230	436	15.83%
Electronics_Product_5932	Electronics	Rs. 12,935,734.47	Rs. 3,731,848.87	276	490	28.85%
Electronics_Product_5361	Electronics	Rs. 12,888,529.86	Rs. 4,036,130.46	224	390	31.32%
Electronics_Product_5301	Electronics	Rs. 12,137,257.01	Rs. 1,981,386.29	257	432	16.32%
Electronics_Product_5696	Electronics	Rs. 11,342,137.80	Rs. 1,713,843.42	241	426	15.11%
Electronics_Product_5855	Electronics	Rs. 11,013,195.82	Rs. 2,138,743.98	293	491	19.42%
Electronics_Product_6099	Electronics	Rs. 10,895,665.30	Rs. 2,311,840.31	254	481	21.22%
Electronics_Product_5785	Electronics	Rs. 10,712,282.01	Rs. 4,129,422.66	255	435	38.55%
Electronics_Product_6130	Electronics	Rs. 10,661,128.30	Rs. 3,623,989.87	246	441	33.99%
Electronics_Product_6167	Electronics	Rs. 10,573,377.25	Rs. 2,188,285.41	249	424	20.70%



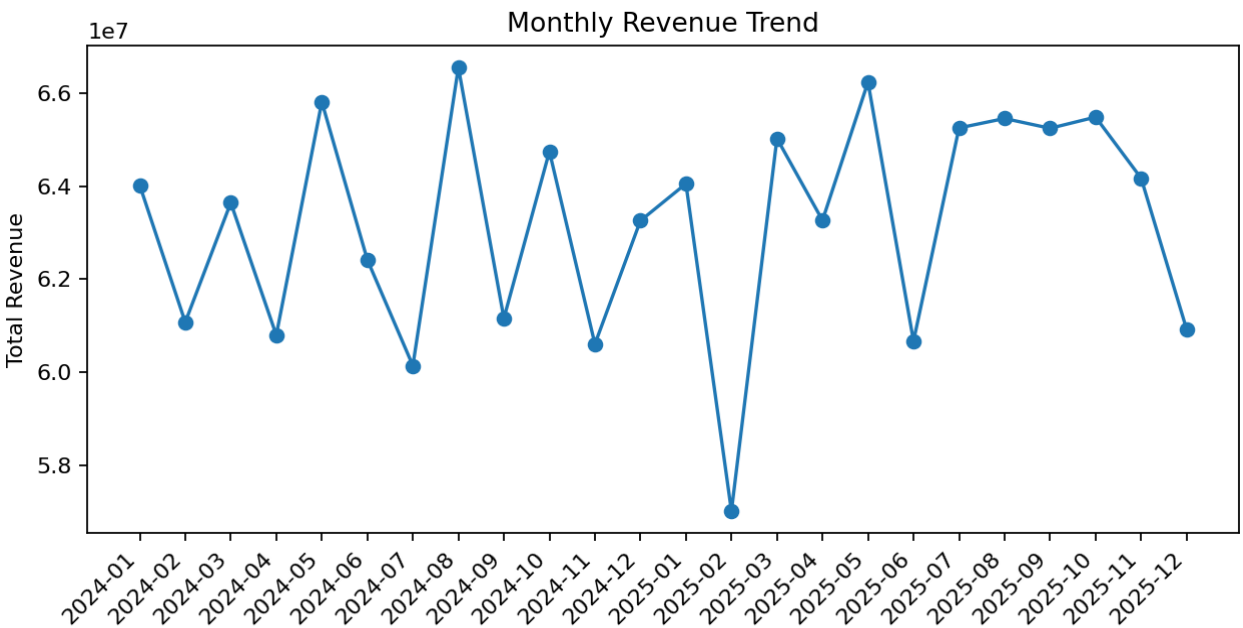
Insight: Top products contribute major revenue concentration. These products should be monitored for stock availability, promotions, and supply planning.

7. Monthly Sales Trend

Monthly Revenue Trend

period	month_name	total_revenue	total_profit	total_orders	avg_order_value
2024-01	Jan	Rs. 64,019,278.14	Rs. 16,965,577.65	12,723	Rs. 5,031.78
2024-02	Feb	Rs. 61,074,100.04	Rs. 16,316,267.15	11,922	Rs. 5,122.81
2024-03	Mar	Rs. 63,643,837.32	Rs. 17,220,926.02	12,483	Rs. 5,098.44
2024-04	Apr	Rs. 60,795,962.48	Rs. 16,018,603.86	12,343	Rs. 4,925.54
2024-05	May	Rs. 65,812,787.28	Rs. 17,719,040.09	12,733	Rs. 5,168.68
2024-06	Jun	Rs. 62,416,883.25	Rs. 16,722,753.54	12,567	Rs. 4,966.73
2024-07	Jul	Rs. 60,139,258.14	Rs. 16,102,234.35	12,541	Rs. 4,795.41
2024-08	Aug	Rs. 66,553,141.42	Rs. 17,778,920.92	12,788	Rs. 5,204.34
2024-09	Sep	Rs. 61,159,371.14	Rs. 16,515,948.10	12,480	Rs. 4,900.59
2024-10	Oct	Rs. 64,746,565.01	Rs. 17,419,001.06	12,960	Rs. 4,995.88
2024-11	Nov	Rs. 60,603,914.69	Rs. 16,228,535.20	12,205	Rs. 4,965.50
2024-12	Dec	Rs. 63,257,004.01	Rs. 16,795,464.14	12,573	Rs. 5,031.18
2025-01	Jan	Rs. 64,053,193.07	Rs. 16,945,932.95	12,779	Rs. 5,012.38
2025-02	Feb	Rs. 57,010,943.04	Rs. 15,110,181.62	11,476	Rs. 4,967.84
2025-03	Mar	Rs. 65,026,356.04	Rs. 17,408,002.50	12,875	Rs. 5,050.59
2025-04	Apr	Rs. 63,265,082.96	Rs. 16,820,810.25	12,377	Rs. 5,111.50
2025-05	May	Rs. 66,238,479.39	Rs. 17,744,680.37	12,757	Rs. 5,192.32
2025-06	Jun	Rs. 60,667,961.01	Rs. 15,932,436.83	12,290	Rs. 4,936.37
2025-07	Jul	Rs. 65,252,577.95	Rs. 17,480,205.79	12,727	Rs. 5,127.10
2025-08	Aug	Rs. 65,455,585.76	Rs. 17,496,548.62	12,730	Rs. 5,141.84
2025-09	Sep	Rs. 65,245,008.50	Rs. 17,297,322.17	12,237	Rs. 5,331.78

period	month_name	total_revenue	total_profit	total_orders	avg_order_value
2025-10	Oct	Rs. 65,485,947.38	Rs. 17,446,197.53	12,604	Rs. 5,195.65
2025-11	Nov	Rs. 64,172,055.88	Rs. 17,133,328.83	12,459	Rs. 5,150.66
2025-12	Dec	Rs. 60,921,963.10	Rs. 16,203,351.90	12,371	Rs. 4,924.58

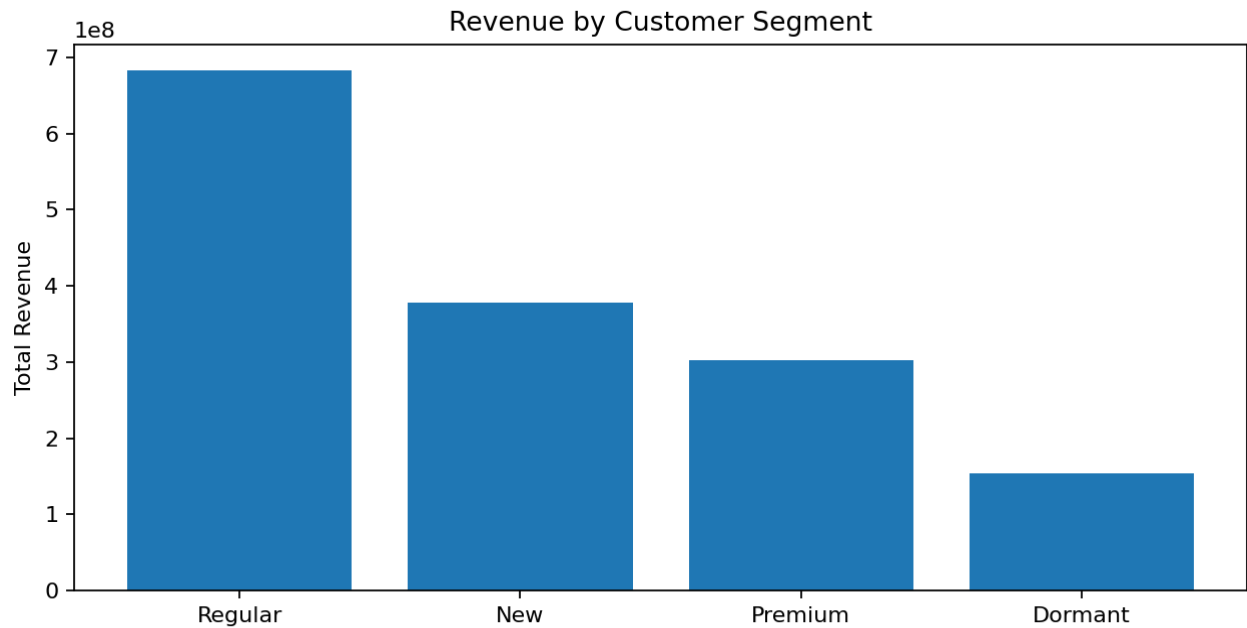


Insight: The highest revenue month was 2024-08 with Rs. 66,553,141.42. Monthly trends help identify seasonality and campaign impact.

8. Customer Segment Analysis

Customer Segment Performance

segment	total_revenue	total_profit	total_orders	total_customers	avg_order_value	profit_margin_percentage
Regular	Rs. 682,897,259.44	Rs. 182,770,094.27	134,630	22,335	Rs. 5,072.40	26.76%
New	Rs. 378,168,867.73	Rs. 100,584,937.19	74,789	12,413	Rs. 5,056.48	26.60%
Premium	Rs. 302,751,359.99	Rs. 80,457,206.22	60,430	10,114	Rs. 5,009.95	26.58%
Dormant	Rs. 153,199,769.84	Rs. 41,010,033.76	30,151	5,025	Rs. 5,081.08	26.77%



Insight: Regular customers contributed the highest revenue. Segment-level analysis helps identify target groups for loyalty programs, premium offers, and reactivation campaigns.

9. City and Store Performance

Top 10 Cities by Revenue

city	state	total_revenue	total_profit	total_orders	total_customers	avg_order_value
Noida	Uttar Pradesh	Rs. 78,992,605.63	Rs. 21,030,263.54	15,401	2,548	Rs. 5,129.06
Nagpur	Maharashtra	Rs. 78,068,462.00	Rs. 20,732,688.16	15,132	2,514	Rs. 5,159.16
Gurugram	Haryana	Rs. 77,427,785.38	Rs. 20,518,313.12	15,137	2,506	Rs. 5,115.13
Bhopal	Madhya Pradesh	Rs. 77,407,930.40	Rs. 20,726,475.57	15,159	2,517	Rs. 5,106.40
Kochi	Kerala	Rs. 77,143,502.96	Rs. 20,571,589.29	14,906	2,475	Rs. 5,175.33
Patna	Bihar	Rs. 77,060,191.36	Rs. 20,643,925.21	15,167	2,536	Rs. 5,080.78
Ahmedabad	Gujarat	Rs. 76,369,903.25	Rs. 20,443,270.22	15,193	2,524	Rs. 5,026.65
Coimbatore	Tamil Nadu	Rs. 76,218,878.06	Rs. 20,138,250.56	15,039	2,486	Rs. 5,068.08
Jaipur	Rajasthan	Rs. 76,114,375.47	Rs. 20,177,902.14	14,499	2,393	Rs. 5,249.63
Chennai	Tamil Nadu	Rs. 76,039,783.34	Rs. 20,358,264.11	15,173	2,510	Rs. 5,011.52

Top 10 Stores by Revenue

store_id	store_name	city	state	total_revenue	total_profit	total_orders	total_quantity_sold	avg_order_value	profit_margin_percentage
2052	Store_2052	Kochi	Kerala	Rs. 13,923,790.27	Rs. 3,802,699.38	2,542	4,458	Rs. 5,477.49	27.31%
2011	Store_2011	Pune	Maharashtra	Rs. 13,852,719.82	Rs. 3,797,553.52	2,559	4,506	Rs. 5,413.33	27.41%
2017	Store_2017	Patna	Bihar	Rs. 13,850,108.56	Rs. 3,824,084.15	2,466	4,379	Rs. 5,616.43	27.61%
2015	Store_2015	Kochi	Kerala	Rs. 13,831,571.96	Rs. 3,608,058.13	2,594	4,531	Rs. 5,332.14	26.09%
2035	Store_2035	Bengaluru	Karnataka	Rs. 13,744,523.21	Rs. 3,707,459.76	2,521	4,511	Rs. 5,452.01	26.97%
2018	Store_2018	Nagpur	Maharashtra	Rs. 13,735,103.22	Rs. 3,639,459.19	2,615	4,592	Rs. 5,252.43	26.50%
2114	Store_2114	Mumbai	Maharashtra	Rs. 13,694,154.33	Rs. 3,700,545.87	2,542	4,506	Rs. 5,387.16	27.02%
2070	Store_2070	Patna	Bihar	Rs. 13,679,544.53	Rs. 3,696,922.13	2,563	4,587	Rs. 5,337.32	27.03%

store_id	store_name	city	state	total_revenue	total_profit	total_orders	total_quantity_sold	avg_order_value	profit_margin_percentage
2045	Store_2045	Bhopal	Madhya Pradesh	Rs. 13,652,080.36	Rs. 3,605,830.71	2,564	4,504	Rs. 5,324.52	26.41%
2118	Store_2118	Bengaluru	Karnataka	Rs. 13,560,981.53	Rs. 3,760,358.29	2,476	4,432	Rs. 5,476.97	27.73%

Insight: Location analysis helps identify strong markets and best-performing stores. These insights can support inventory distribution, staffing, and regional campaigns.

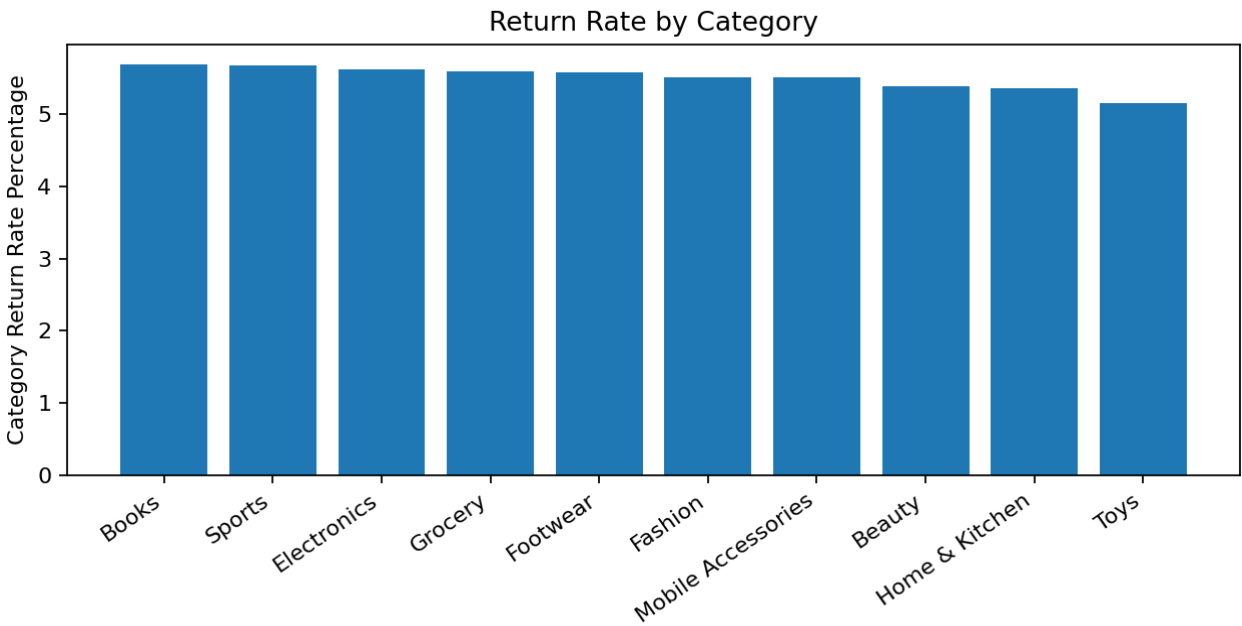
10. Returns Analysis

Return Summary

total_returns	returned_orders	return_rate_percentage	returned_order_revenue
16,513	16,513	5.50%	Rs. 85,178,796.38

Returns by Product Category

category	total_returns	total_orders	returned_revenue	category_return_rate_percentage
Books	1,691	29,744	Rs. 1,294,345.31	5.69%
Sports	1,769	31,173	Rs. 7,373,347.54	5.67%
Electronics	1,668	29,651	Rs. 48,132,806.66	5.63%
Grocery	1,591	28,418	Rs. 1,549,007.30	5.60%
Footwear	1,676	30,023	Rs. 5,773,412.02	5.58%
Fashion	1,634	29,660	Rs. 4,639,714.07	5.51%
Mobile Accessories	1,707	30,999	Rs. 2,198,126.95	5.51%
Beauty	1,434	26,630	Rs. 2,015,012.45	5.38%
Home & Kitchen	1,514	28,240	Rs. 8,894,342.56	5.36%
Toys	1,829	35,462	Rs. 3,308,681.52	5.16%

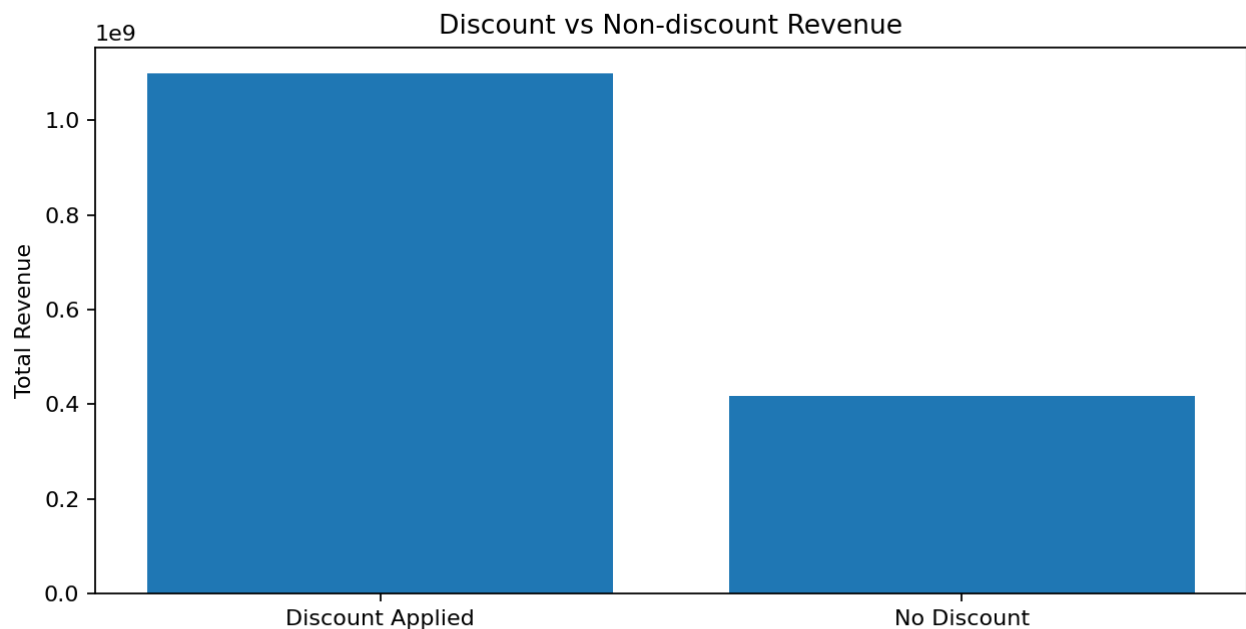


Insight: The overall return rate is 5.50%. Category-level return analysis can reveal product quality, sizing, logistics, or customer expectation issues.

11. Discount and Payment Analysis

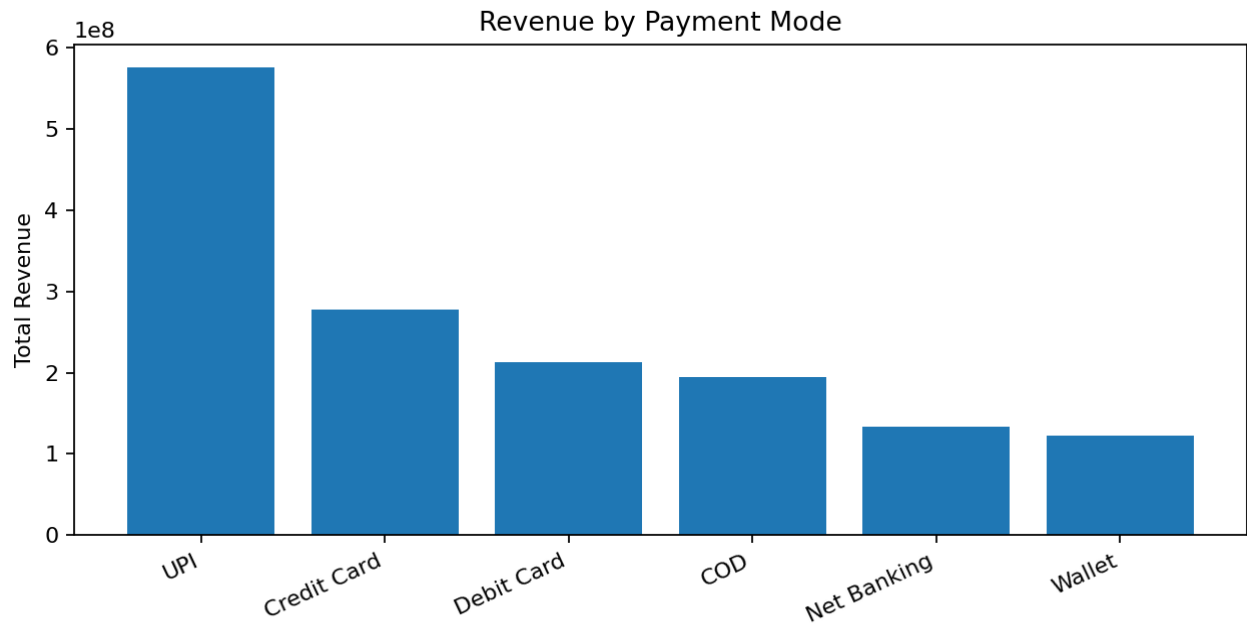
Discount Impact Analysis

discount_status	total_revenue	total_profit	total_orders	avg_order_value	profit_margin_percentage
Discount Applied	Rs. 1,099,090,843.79	Rs. 263,614,782.41	225,228	Rs. 4,879.90	23.98%
No Discount	Rs. 417,926,413.21	Rs. 141,207,489.03	74,772	Rs. 5,589.34	33.79%



Payment Mode Analysis

payment_mode	total_revenue	total_orders	avg_order_value	order_percentage
UPI	Rs. 575,074,980.82	114,273	Rs. 5,032.47	38.09%
Credit Card	Rs. 277,844,095.62	53,929	Rs. 5,152.04	17.98%
Debit Card	Rs. 213,153,206.26	42,028	Rs. 5,071.70	14.01%
COD	Rs. 194,597,529.73	38,683	Rs. 5,030.57	12.89%
Net Banking	Rs. 133,324,578.97	26,985	Rs. 4,940.69	9.00%
Wallet	Rs. 123,022,865.60	24,102	Rs. 5,104.26	8.03%



Insight: Discount analysis compares order behavior with and without discounts. Payment mode analysis shows that UPI is the dominant payment method by revenue and order count.

12. Repeat Customer Analysis

Repeat vs One-time Customers

customer_type	total_customers
One-time Customer	728
Repeat Customer	49,159

Insight: Most customers are repeat customers. This indicates strong retention behavior in the dataset and supports building customer loyalty metrics in Power BI.